

ROOTING FOR KEPLER

*a comparative benchmark of
Newton-type root finders and starting guesses*

An empirical study of classical, domain-specialised, and
2024–2025 with-memory Newton-type variants applied to
Kepler’s equation, evaluated on RadVel’s radial-velocity fitting pipeline.

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1 MOTIVATION

Radial-velocity orbit fitting reduces, at its numerical core, to a single repeated task: solving the transcendental Kepler equation

$$f(E) = E - e \sin E - M = 0$$

for the eccentric anomaly E , given a mean anomaly M and eccentricity e . A fitting pipeline such as RADVEL calls this solver on the order of 10^5 – 10^7 times over the course of a single MCMC run, so the choice of root finder is not a cosmetic detail — it is one of the dominant costs in the fit, and its accuracy floor propagates directly into the recovered orbital parameters.

RADVEL itself uses **Danby's** higher-order iterative method rather than an ordinary Newton–Raphson update, which already signals that the "obvious" solver is not automatically the best one for this equation. At the same time, the general numerical-analysis literature on Newton-type iteration has continued to move: 2024–2025 work on **with-memory** multipoint schemes reports polynomial convergence orders in excess of 10, obtained by recycling already-computed function values across iterations rather than by paying for new derivative evaluations.

WHY THIS IS NOT A SETTLED QUESTION

None of the recent with-memory papers were written with Kepler's equation in mind, and Kepler's equation has a cost structure — one transcendental call can cheaply return *both* $\sin E$ and $\cos E$ — that generic efficiency-index comparisons ($E = \rho^{1/\gamma}$) do not capture. Whether a higher formal convergence order actually pays off once this Kepler-specific evaluation cost is accounted for, and across the (e, M) operating range that real radial-velocity fits encounter, is an open, testable question rather than a foregone conclusion.

2 SCOPE STATEMENT

The purpose of this project is to build a controlled, Kepler-specific benchmark of five root-finding variants and one alternative starting-guess strategy, and to measure — not assume — how they compare.

WE ARE EXPLICITLY *not* SETTING OUT TO PROVE THAT THE NEWER WITH-MEMORY NEWTON-TYPE VARIANTS OUTPERFORM, OR UNDERPERFORM, THE CLASSICAL SOLVERS ALREADY USED FOR KEPLER'S EQUATION. The project is designed as an experiment whose outcome is not fixed in advance: the newer variants may win on some axes (order, iteration count) and lose on others (per-iteration cost, robustness at high eccentricity), and part of the deliverable is characterising exactly where that trade-off falls.

ADDRESSING SUPERVISOR FEEDBACK

An earlier draft of this proposal was scoped narrowly, around substituting an ordinary Newton–Raphson update for the Danby iteration currently used in RADVEL. Supervisory review identified that this framing was too restrictive. The standard Newton–Raphson method is unlikely to be the right replacement for a domain-tuned higher-order scheme, and a study built around a single alternative could not establish whether any observed difference was attributable to the iteration itself. The revisions below were made in response, and they account for the present structure of Sections 3 and 4.

Feedback	Revision	Section
Do not limit the study to standard Newton–Raphson.	Candidate set widened to five solvers across three categories.	3
Explore a range of alternatives, not one substitution.	Two with-memory schemes (2024–2025) added.	3.2
Retain the baseline so results stay interpretable.	Newton–Raphson kept as anchor; category sizes balanced.	3.1
Do not presuppose that the newer methods are superior.	Scope statement fixes the design, not the expected result.	2, 6

The net effect of these revisions is a change of research question. The study no longer asks whether one specific solver can replace another. It asks under which conditions of eccentricity, mean anomaly, and Kepler-specific evaluation cost each class of Newton-type method is preferable. This is a question whose answer is not determined in advance by the choice of methods.

3 CANDIDATE METHODS

Five solvers are selected to span three categories that are each represented by exactly two members, plus one initial-guess strategy that is evaluated orthogonally to all of them.

3.1 SELECTION RATIONALE

- **Baseline:** ordinary Newton–Raphson, to anchor every comparison against the textbook method.
- **Established Kepler-specific solvers:** two variants that are already used or benchmarked against in the Kepler-equation literature, representing the current practical standard rather than a strawman.
- **Recent with-memory variants (2024–2025):** two multipoint Newton-type schemes that raise convergence order via Hermite-interpolated self-accelerating parameters, at no additional evaluation points per iteration.
- **Initial-guess layer:** a symbolic-regression-derived starting formula, applied in front of every iterative solver above, isolated as its own experimental factor.

3.2 METHOD TABLE

Category	Method	Order	Role in the benchmark
Baseline	Newton–Raphson	2	Textbook anchor; one evaluation point, full sin, cos pair per iteration.
Established Kepler solvers	Danby’s method	4	RadVel’s production solver; already domain-tuned to this equation.
	Markley’s method	—	Closed-form / near-analytic quartic solver widely cited as a Kepler-specific standard.
With-memory (2024–2025)	NWM9 (Mittal, Panday & Jäntschi, 2024)	8.8989	Single self-accelerating parameter on an optimal 8th-order base; controlled, lower-order sibling of NWM11.
	NWM11 (Mittal, Panday, Jäntschi & Bolundut, 2025)	10.7446	Bi-parametric variant; primary "recent" candidate — 3 evaluation points, only 1 full sin, cos pair per iteration.
Guess layer	Napier’s guess (2024, arXiv:2411.15374)	n/a	Symbolic-regression starting formula; drop-in replacement for the simple/canonical/RadVel starts, used in front of every solver above.

Two established, two new, and one baseline gives every iterative solver a same-size comparison group, while the guess-layer factor is deliberately kept independent so that "better start" and "better iteration" are never conflated in the results.

4 EVALUATION FRAMEWORK

4.1 KEPLER-SPECIFIC COST ACCOUNTING

Raw convergence order is not, by itself, a fair currency for comparing these methods on this particular equation. A single `sincos` call returns $\sin E$ and $\cos E$ together at roughly the cost of one transcendental evaluation, so a scheme that needs both quantities at one point is markedly cheaper than one that needs sin and cos at two different points. Every method above will therefore be costed by:

1. the number of *distinct* evaluation points used per iteration;
2. how many of those points require a full sin, cos pair versus sin (or cos) alone; and
3. whether any derivative at a new point is synthesised from divided-difference / Hermite machinery instead of a fresh transcendental call.

4.2 BENCHMARK GRID

Each solver–guess combination will be run over a broad (e, M) grid, including the ordinary operating region of typical radial-velocity fits and the classical pathological corner — high eccentricity, M near zero — where Newton-type iteration for Kepler’s equation is known to be most fragile.

4.3 METRICS

- **Empirical convergence order**, measured directly from the residual sequence rather than taken on faith from the source paper.
- **Iterations to tolerance** and **wall-clock time** per solve, and per full RadVel fit.

- **Residual and reference-solution error** against a high-precision root.
- **Robustness:** failure or non-convergence rate across the (e, M) grid.
- **Downstream effect:** propagation of solver error through $M \rightarrow E \rightarrow v \rightarrow v_r$ into the fitted parameters P , e , and K , and comparison against RadVel’s MCMC posterior scale.

5 WORK PLAN

1. **Implementation.** Re-derive and implement all five solvers and the Napier guess in a common scalar-arithmetic framework, instrumented to record evaluation-point-level cost, not just iteration count.
2. **Verification.** Validate each implementation against its source paper’s reported convergence order on a small controlled test set before running the full grid.
3. **Grid benchmark.** Run every solver–guess combination across the (e, M) grid; record all metrics in Section 4.3.
4. **Error propagation study.** Vary solver tolerance and trace the resulting error through to refit orbital parameters on a real radial-velocity dataset.
5. **Monte Carlo uncertainty comparison.** Generate noisy realisations of the same dataset, refit, and compare measurement-noise-driven parameter spread against solver-induced shifts and RadVel’s MCMC posterior.
6. **Synthesis.** Report where each method category wins, loses, or ties, once Kepler-specific evaluation cost is accounted for.

6 EXPECTED OUTCOMES

- A reusable, cost-instrumented benchmark harness comparing classical, domain-specialised, and 2024–2025 with-memory Newton-type solvers on Kepler’s equation specifically.
- An empirical answer — not an assumed one — to whether the higher formal order of recent with-memory schemes survives contact with Kepler’s specific $\sin, \cos$ cost structure and RadVel’s real operating range of (e, M) .
- A quantified picture of how solver-level numerical error propagates into fitted orbital parameters, set against RadVel’s own MCMC uncertainty for context.

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